1.What is a Program?

ans:- A program is a set of instructions that tells a computer what to do.
For example:

•A calculator app is a program that tells the computer how to add, subtract, multiply, and divide.
•A game is a program that tells the computer how characters move and how the game works.
•Even apps like WhatsApp and Instagram run using programs.

*LAB EXERCISE: Write a simple "Hello World" program in two different programming languages ofyour choice. Compare the structure and syntax.

```
Ans:-#include<stdio.h>
#include<conio.h>
void main()
{
  printf("Hello World");
  getch();
}
```

2.What is Programming?

Ans:-Programming means giving instructions to a computer so it can do tasks for you.

For example:
•If you want a calculator app → you program it.
•If you want a game → you program it.
•If you want a website → you program it.
These instructions are written using programming languages like:
Python
Java
C++

*THEORY EXERCISE: What are the key steps involved in the programming process?

Ans:-1.Problem Understanding
•What is the goal?
•What inputs will you get?
•What output is expected?

2.Documentation
•How the program works
•How to use it
•Important parts of the code

3.Testing and Debugging
•Does it work correctly?
•Are there any errors (bugs)?
Fix any mistakes found during testing.

3.Types of Programming Languages

Ans:-1.Low-Level Languages
• Machine Language
• Assembly Language

2.High-Level Languages
These are easy for humans to read and write.They use simple English-like words.
Examples:
•Python
•Java
•C++

3.Object-Oriented Programming (OOP) Languages
Examples:
•Java
•C++

Used mainly for small tasks, automation, or web development.

*THEORY EXERCISE: What are the main differences between high-level and low-level programminglanguages?

*Low-Level = Close to computer
•Hard for humans to understand
•Close to computer hardware
•Very fast execution

*High-Level = Easy for humans
•Easy for humans to read and write
•Use simple English-like words

4.World Wide Web & How Internet Works

Ans:-• A server is where websites are stored, and it works a lot like yourcomputer's hard drive.
• Once the request arrives, the server retrieves the website and sendsthe correct data back to your computer.

1.Client & Servers
1. Client Program
• Running on end host
• Requests Services
• E.g. Web Browser

2.Server Program
• Running on end host
• Provides Services
• E.g. Webserver

*THEORY EXERCISE: Describe the roles of the client and server in web communication.
Ans:-In web communication, client and server work together like a customer and a shop.
1.Client:-
•A web browser like Google Chrome
•A mobile browser like Mozilla Firefox
•Apps on your phone

2.Server:-
The server is a powerful computer that stores websites and data.
•Receives the request from the client
•Processes the request
•Sends back the required data (like a webpage)

5.Network Layers on Client and Server

Ans:-
1.Physical Layer
•On client: Your phone's Wi-Fi or mobile signal
•On server: Data center cables

2.Data Link Layer
•Uses MAC address
•Controls basic error checking

3.Network Layer
•Uses IP address
•Example: Internet routing

4.Transport Layer
•TCP (safe and reliable)
•UDP (fast but no guarantee)

5.Session Layer
6.Presentation Layer
7.Application Layer
•Web browser
•Email app
•WhatsApp

*THEORY EXERCISE: Explain the function of the TCP/IP model and its layers.
Ans:-
The TCP/IP model is a way to explain how computers send and receive data over the internet.
It breaks the communication process into 4 simple layers, and each layer has a specific job.
1.Application Layer
2.Transport Layer
3.Internet Layer
4.Network Access Layer

6.Client and Servers

• Client
A client is a device or app that requests something.
Examples:
•A web browser like Google Chrome
•Your phone or computer

•Server
A server is a powerful computer that stores information and sends it when requested.
Examples:
•Google servers store search data
•Instagram servers store photos and videos

*THEORY EXERCISE: Explain Client Server Communication
Ans:-
Client–Server communication is the way computers talk to each other over the internet.
•What is a Client?
A client is a device or program that requests something. Example:
•A web browser like Google Chrome
•A mobile app like WhatsApp

When you search for something or send a message, your device acts as a client.

•What is a Server?
A server is a powerful computer that stores data and responds to requests from clients.

For example:
•When you open YouTube, your phone (client) asks YouTube's server for a video.
•The server sends the video back to your phone

*How Client–Server Communication Works (Step-by-Step)
1.Request – The client sends a request to the server.(Example: "Show me this website.")

2.Processing – The server receives the request and processes it.

3.Response – The server sends back the required data.

4.Display – The client shows the result to the user.

7.Types of Internet Connections

Ans:-
1.Fiber Optic Internet
Uses special glass cables that send data as light.
•Very fast
• Very reliable
• Can be more expensive

2.DSL (Digital Subscriber Line)
Uses old telephone lines.
• Faster than dial-up
• Affordable
•Speed depends on distance from provider
•Best for: Basic home use

3.Cable Internet
Uses the same cables as cable TV.
•Fast speeds
• Speed may slow down when many people use it
•Best for: Families and regular streaming

4.Satellite Internet
Uses satellites in space to provide internet.
•Available in rural/remote areas
• Can be slower with delay (lag)
•Weather can affect connection
•Best for: Areas without wired internet

5.Dial-Up
Very old type of internet using phone lines.
• Very slow
•Cannot use phone and internet at the same time
•Mostly outdated today

*THEORY EXERCISE: How does broadband differ from fiber-optic internet?
Ans:-
1.Broadband

Broadband is a general term.
It means high-speed internet that is always connected.

Broadband can include:
•DSL
•Cable
•Fiber-optic
•Satellite
•Mobile internet

2.Fiber-Optic Internet
Fiber-optic internet is a type of broadband.
It uses special glass cables that send data as light.

•Very fast
•Very reliable
•Handles heavy use (gaming, streaming, video calls) very well

8.Protocols

Ans:-

A protocol is a set of rules that tells computers how to talk to each other.

Just like people follow rules when talking (one person speaks, the other listens), computers also need rules to send and receive information correctly.

Example:

•Think of sending a letter:
•You write the message.
•You put the correct address.
•You follow postal rules.

Computers do the same thing using protocols.

•Common Internet Protocols:

HTTP
Used to load websites.

HTTPS
Same as HTTP but more secure.

FTP
Used to send files between computers.

SMTP
Used to send emails.

***THEORY EXERCISE: What are the differences between HTTP and HTTPS protocols? Application Security**

HTTP:-
•Not secure
•Data is sent in plain text
•Anyone can read the data
•Starts with http://

HTTPS:-
•Secure
•Data is encrypted (hidden)
•Hackers cannot easily read the data
•Starts with https://

Application Security means protecting software and apps from hackers and attacks.

Examples of Application Security:

•Using HTTPS
•Strong passwords
•Login authentication
•Firewalls
•Regular updates

9.Application Security

Ans:-
Application Security means protecting software applications from hackers and attacks.

*Why is Application Security Important?

•Protects user data
•Prevents hacking
•Keeps passwords safe
•Stops viruses and malware
•Builds trust with user's

*How Does Application Security Work?

•Strong passwords
•Encryption (scrambling data so hackers can't read it)
•Authentication (checking who you are)
•Authorization (deciding what you are allowed to do)
•Security testing (finding and fixing weaknesses)

•When you use apps like Facebook, Instagram, or Amazon, your personal data (like passwords, photos, card details) must be kept safe.Application security makes sure no one can steal, change, or damage that information.

***THEORY EXERCISE: What is the role of encryption in securing applications?**

Ans:-
Encryption is like putting your information inside a secret lock box so that only the right person can open it.

Encryption scrambles this information into a secret code while it is being sent over the internet.

Encryption:-

1.Protects Sensitive Data
Keeps passwords, bank details, and personal data safe.

2.Prevents Hackers from Reading Data
Even if data is intercepted, it cannot be understood.

3.Builds Trust
Users feel safe using apps that protect their information.

4.Keeps Data Safe During Transfer
For example, when you see HTTPS in a website address, it means encryption is protecting the connection

10.Software Applications and Its Types

Ans:-

-It is a type of software application that helps in the automation of the task based on the Users Input.

-There are the different application which helps us in our daily life to process our instructions based on certain rules and regulations.

-The user may use the computer for browsing the internet, accessing to emailservice, attending meetings, and playing games.

- Different high-level languages are used to build application software.

*Types of Application Software:-

● Application software
● System software
● Driver software
● Middleware
● Programming software

***THEORY EXERCISE: What is the difference between system software and application software?**

Ans:-
System Software